

Multifunctional Fireguard Bot

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Abstract—This paper presents the design and implementation of a Multifunctional Fireguard Bot, an ESP32-based intelligent robotic system for early fire detection and response. The robot integrates multiple features including line following using IR sensors, obstacle avoidance, flame detection with a dedicated flame sensor, and wireless Wi-Fi control via smartphone for remote operation. Equipped with an ESP32 Camera Module, it also provides real-time visual monitoring. Upon detecting fire, the system triggers an immediate buzzer alarm and supports autonomous navigation along predefined paths while avoiding obstacles. The robot operates in both autonomous and manual modes, making it highly adaptable for different scenarios. Experimental results demonstrate reliable line following, accurate obstacle detection, responsive fire sensing, and stable wireless control. This low-cost, multifunctional robotic solution aims to enhance safety in homes, industries, and smart buildings by providing early hazard detection and rapid response capabilities. The developed prototype successfully showcases the potential of ESP32-powered robotics in real-world fire safety and surveillance applications.

Index Terms—ESP32, Fireguard Bot, Flame Detection, IR Sensor, Line Following Robot, Obstacle Avoidance, Wireless Control, IoT Robotics, Safety System

I. INTRODUCTION

Fire accidents remain major threats to life, property, and industrial infrastructure. Traditional detection systems often suffer from delayed response due to reliance on fixed sensors and human intervention. Most existing fire-fighting robots are limited to single functions and lack integrated autonomous navigation and remote control capabilities. This project presents the Multifunctional Fireguard Bot, an ESP32-based intelligent robotic system that integrates line following, obstacle avoidance, flame detection, and Wi-Fi-based remote control. Designed as an intelligent sentinel, the robot aims to provide early hazard detection and rapid response in homes, industries, and smart environments.

The main objectives of this project are:

- 1) To develop a line-following robot capable of accurate movement along predefined paths using IR sensors,
- 2) To implement obstacle avoidance for safe and smooth navigation,
- 3) To enable real-time fire detection using a flame sensor with immediate buzzer alert,
- 4) To provide wireless Wi-Fi control through a smartphone for remote manual operation.

A. Background and Rationale

With the increasing demand for automation and safety in smart cities and industries, there is a growing need for versatile robotic solutions that can operate both autonomously and under human supervision. The integration of multiple features — line following, obstacle avoidance, fire detection, and remote control — into a single low-cost platform using the powerful ESP32 microcontroller addresses the limitations of existing single-function robots.

B. Scope

This project develops a prototype of the Multifunctional Fireguard Bot as an intelligent safety robot powered by the ESP32 Camera Module. The work mainly concentrates on building a mobile robotic platform that can move autonomously while detecting fire hazards and allowing remote operation. The prototype incorporates the following core functionalities:

- Autonomous navigation through line following and obstacle avoidance,
- Early fire detection using a flame sensor accompanied by an audible buzzer alert,
- Wireless remote control capability using Wi-Fi and smartphone,
- Flexible operation in both autonomous and manual modes.

Limitations of the current version include the absence of automatic fire extinguishing capability and advanced vision-based fire source tracking. The developed system serves as a proof-of-concept to showcase how a single low-cost robot can combine mobility, sensing, and remote control for fire safety applications.

II. RELATED WORK

A significant amount of research has been conducted on fire detection robots and autonomous navigation systems. Most previous implementations using Arduino or ESP32 platforms typically focus on individual features such as basic flame sensing, simple obstacle avoidance, or basic line following [1]. Many ESP32-based projects have successfully demonstrated remote control through Bluetooth or Wi-Fi and basic alerting mechanisms.

Studies have shown that IR sensors provide reliable performance in line following applications [2], while

ultrasonic sensors are commonly adopted for obstacle detection in mobile robots. However, the majority of existing low-cost solutions suffer from a major limitation — they generally support only one or two functionalities and fail to integrate smooth autonomous navigation with hazard detection and wireless control in a single system.

This project addresses these gaps by developing the Multifunctional Fireguard Bot, which successfully combines line following, obstacle avoidance, real-time flame detection with buzzer alert, and Wi-Fi-based smartphone control into one unified platform using the ESP32.

III. METHODOLOGY

A. Overview

The Multifunctional Fireguard Bot consists of three main subsystems:

The system consists of:

- **Sensing Unit :** IR sensors for line following, triple flame sensors for fire detection.
- **Actuation Unit :** DC motors with L298N driver
- **Control and Communication Unit :** Dual-controller architecture using ESP32-CAM (Primary) and Arduino Uno (Secondary) with Wi-Fi web interface.

B. System Architecture

The ESP32 collects data from sensors and controls a pump and servo. Blynk enables monitoring and control.

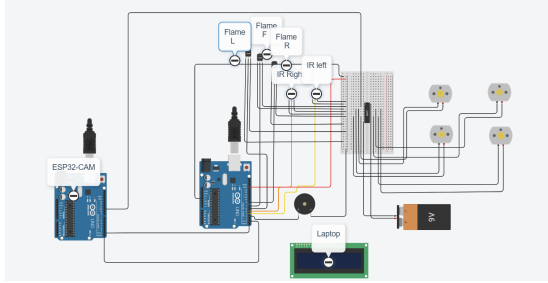


Fig. 1. System architecture of the Multifunctional Fire Guard Bot

TABLE I
PRIMARY HARDWARE COMPONENTS

Component	Role	Qty
ESP32-CAM	Primary controller, Wi-Fi Camera	1
Arduino Uno	Secondary controller, sensors actuators	1
L298N Motor Driver	Motor control (PWM) & humidity	1
DC Gear Motors	Locomotion	4
Flame Sensor	Fire detection	3
IR Tracking Sensor	Line following	2
Active Buzzer	Fire alert	1
18650 Li-ion	Power supply	3

C. Communication Architecture

The system uses a Primary-Secondary communication model. The Arduino Uno sends high-level movement and fire detection commands to the ESP32-CAM via SoftwareSerial. The ESP32-CAM hosts a SoftAP Wi-Fi network (SSID: FireBot) and serves a responsive web interface for manual control and mode selection. For AI-enhanced navigation, a laptop running YOLOv11 processes the video stream and sends HTTP commands back to the ESP32-CAM.

D. Control Logic and Operation Modes

The system evaluates sensor inputs and operates based on the selected mode:

- **Line Follower Mode:** Dual IR sensors detect the black line and send appropriate movement commands (Forward, Left, Right).
- **Firefighter Mode:** Triple flame sensors detect fire presence and direction (Left, Center, Right). Upon detection, the robot stops, activates the buzzer.
- **Manual Mode:** User controls the robot through the web dashboard with directional buttons and speed control (PWM 50–255).
- **YOLO AI Mode:** Real-time object detection via YOLOv11 on a connected laptop for obstacle avoidance.

Safety Features:

- Emergency stop button instantly halts all operations.

E. Control and Operation Sequence

The automated operation of the Multifunctional Fireguard Bot follows a structured process as shown in Fig.2. After power-up, the system initializes the ESP32-CAM, Arduino Uno, and sensors, and the ESP32-CAM creates a Wi-Fi SoftAP with SSID FireBot. The user connects to the robot's Wi-Fi network and accesses the web interface at <http://192.168.4.1> to select the desired operating mode.

The system supports four operational modes:

- In Manual Mode, the user controls the robot using the D-Pad on the web interface. Commands are sent via HTTP GET requests to the ESP32-CAM, which drives the L298N motor driver using PWM signals.
- In Line Follower Mode, the Arduino Uno continuously reads the dual IR sensors (D4 and D5). Based on sensor readings, it sends appropriate movement commands (Forward, Left, or Right) to the ESP32-CAM via serial communication.
- In Fire Mode, the Arduino Uno polls the triple flame sensors (D6, D7, D8). Upon flame detection, the robot stops, activates the buzzer (D3), and remains in alert state until the flame is cleared.
- In YOLO AI Mode, a connected laptop fetches the MJPEG video stream from the ESP32-CAM, performs real-time object detection using YOLOv11, and sends HTTP commands back to the ESP32-CAM to avoid obstacles.

Throughout all modes, the system continuously checks for the emergency stop command. When triggered, all motors and operations are immediately halted. The priority-based decision logic ensures safety-critical actions (emergency stop and fire detection) take precedence. This real-time cycle of sensing, decision-making, communication, and actuation enables reliable autonomous and manual operation of the robot.

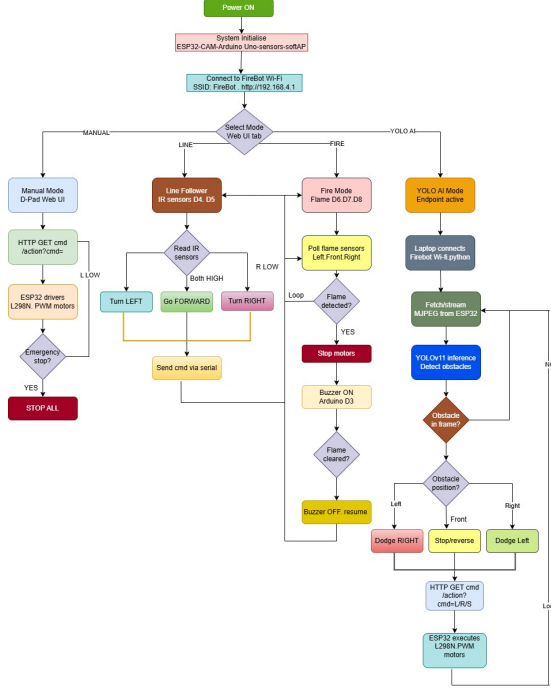


Fig. 2. Flowchart of the control and operation

IV. IMPLEMENTATION

A. Microcontroller Programming

The firmware for both microcontrollers was developed using the Arduino IDE. The ESP32-CAM code handles Wi-Fi SoftAP creation, web server hosting, camera streaming, and motor control through the L298N driver using PWM signals. The Arduino Uno code manages real-time sensor acquisition and buzzer control using the `SoftwareSerial.h` library.

At startup, both controllers are initialized, serial communication is established at 9600 baud, and the ESP32-CAM begins broadcasting the Wi-Fi network with SSID FireBot.

B. Sensor Integration

Each sensing module was individually tested and calibrated for reliable performance:

- **IR Tracking Sensors:** Placed at the front bottom of the chassis for accurate line detection on black lines. Threshold values were tuned according to surface conditions.
- **Flame Sensors:** Three flame sensors (Left, Front, Right) were used to detect fire and determine its approximate direction.

- **Camera:** The built-in camera of ESP32-CAM was configured for real-time video streaming to support the YOLOv11-based obstacle avoidance system.

Sensor data from the Arduino Uno is continuously shared with the ESP32-CAM for coordinated decision making.

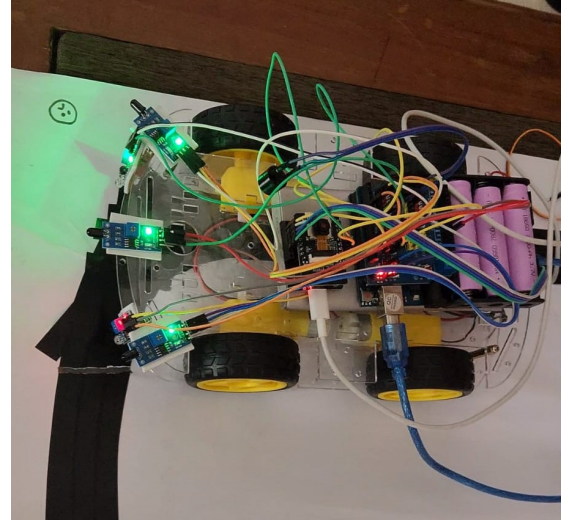


Fig. 3. System prototype

C. Actuator Control

- **Motor Control:** The L298N motor driver is controlled by the ESP32-CAM using four GPIO pins with PWM capability for variable speed control (50–255).
- **Alert System:** Upon fire detection by any flame sensor, an active buzzer is immediately triggered to provide audible alert.

D. Web Interface and Manual Control

A responsive web dashboard was developed using HTML, CSS, and JavaScript. The interface provides the following features:

- Mode selection (Manual, Line Follower, Firefighter, YOLO AI)
- Virtual D-Pad for manual movement control
- Speed control slider
- Global emergency stop button with full-screen lockdown overlay

The dashboard is accessible at <http://192.168.4.1> when connected to the robot's Wi-Fi network.

E. YOLOv11 AI Integration

A Python script running on a connected laptop uses OpenCV and Ultralytics YOLOv11 library to process the live video stream from the ESP32-CAM. Detected obstacles are analyzed in real-time, and appropriate movement commands are sent back to the ESP32-CAM via HTTP requests to enable autonomous obstacle avoidance.

V. RESULTS AND DISCUSSION

A. Operational Results

The Multifunctional Fireguard Bot was successfully implemented and tested under various conditions. The system demonstrated the following key functionalities:

- Reliable line following using dual IR sensors on predefined black line paths.
- Accurate fire detection with triple flame sensors and immediate buzzer activation.
- Responsive manual control through the web interface with smooth motor movement and speed adjustment.
- Real-time video streaming from the ESP32-CAM.
- Effective obstacle avoidance using YOLOv11 running on a laptop.
- Instant response to the emergency stop command.

The dual-controller architecture (ESP32-CAM + Arduino Uno) provided stable performance with good coordination between sensing and actuation.

B. Limitations

During testing, the system exhibited some limitations:

- Flame sensor performance is affected by ambient light and distance.
- Line following accuracy depends on line thickness and surface contrast.
- Wi-Fi range limits the remote control distance (effective up to 10–15 meters).
- YOLOv11-based obstacle avoidance requires a separate laptop for processing.
- Battery life is relatively short during continuous operation.

C. Performance Evaluation

The robot showed fast response time for fire detection and buzzer activation. Manual control via web interface was responsive with minimal latency within the Wi-Fi range. Line following worked reliably on clear paths. The overall system demonstrated good integration of multiple features, proving the effectiveness of the Primary-Secondary controller design for multifunctional robotic applications.

D. System Learnings

Through the design, implementation, and testing of the Multifunctional Fireguard Bot, several key lessons were identified:

- Proper calibration of IR and flame sensors is crucial to achieve consistent and reliable performance under varying environmental conditions.
- The Primary-Secondary controller architecture (ESP32-CAM + Arduino Uno) is highly effective for separating real-time sensor processing from wireless communication tasks.
- Maintaining a stable power supply is essential to prevent unexpected resets and erratic sensor readings.
- Using non-blocking code and priority-based decision logic significantly improves system responsiveness and reliability.

- Integrating multiple features (line following, fire detection, manual control, and AI) in one platform increases complexity, making modular testing and debugging very important.

These insights will guide future improvements and optimizations in similar IoT-based robotic systems.

VI. CONCLUSION AND FUTURE WORK

This project successfully developed a Multifunctional Fireguard Bot that integrates line following, fire detection with buzzer alert, manual web-based control, and YOLOv11-based obstacle avoidance capabilities. The dual-controller architecture using ESP32-CAM and Arduino Uno proved effective in handling both real-time sensing and wireless communication tasks.

The system demonstrates a practical and low-cost solution for autonomous and remotely controllable robotic applications, particularly useful for safety and surveillance purposes. Through this work, the team gained valuable hands-on experience in microcontroller interfacing, sensor integration, wireless communication, and AI-assisted robotics.

Future Work

Several enhancements are proposed for future development:

- Integration of gas sensors (e.g., MQ-2 or MQ-135) for gas leakage detection.
- Implementation of onboard edge AI to eliminate the need for a separate laptop for YOLOv11 processing.
- Addition of GPS module for autonomous navigation and mapping.
- Improvement of battery management system for longer operational time.
- Development of a mobile application for better user experience and extended range.

The Multifunctional Fireguard Bot serves as a strong foundation for more advanced safety and assistive robotic systems.

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